

Green Fleet Optimization Report

PS-138 — Quantum-Inspired Fuel Consumption Prediction and Green Fleet Optimization

1. Executive Summary

A fleet of 16 vessels was evaluated against 4 route demands using the **QGA** optimiser. The recommended plan consumes **2,913 t** of fuel at a total cost of **\$4,356,096**, emitting **7,012 t** CO2e on a lifecycle basis, while meeting **100.0%** of cargo demand at **100.0%** schedule reliability. Constraint status: **feasible**.

2. Fleet Information

Vessel class	Count	Total DWT	Mean DWT
Capesize	3	379,200	126,400
Panamax	6	517,746	86,291
Supramax	5	254,248	50,850
Handysize	2	52,999	26,500

3. Input Scenario

Route	Distance (nm)	Cargo demand (t)	Deadline (h)	Weather
Port Hedland - Qingdao	3,600	240,000	340	MODERATE
Tubarao - Rotterdam	5,100	150,000	470	ROUGH
Paradip - Singapore	2,100	90,000	215	CALM
Richards Bay - Paradip	4,600	120,000	430	MODERATE

Carbon price \$120/t. Emission ceiling: 9000.0. Maximum tolerated delay 24 h.

4. Fuel Consumption Prediction

ML model **GradientBoosting** predicted **371.4 t** for the reference voyage (95% interval 198.3 to 544.5 t).

Model	MAE (t)	RMSE (t)	R2	MAPE (%)
RandomForest	67.46	118.98	0.9479	11.99
GradientBoosting	46.19	88.32	0.9713	7.79

Promoted model: **GradientBoosting**, selected by held-out RMSE on 1,200 unseen voyages.

5. Optimization Methodology

Discrete decisions (vessel deployment, route assignment, fuel selection) are solved with a Quantum-Inspired Genetic Algorithm in which each gene is a vector of probability amplitudes evolved by a quantum rotation gate and collapsed by observation. Continuous decisions (cruising speed per vessel) are solved with Quantum-Inspired Particle Swarm Optimization, which replaces velocity with a delta-potential-well position collapse. Both execute on classical hardware; no quantum computer is used at any point.

6. Selected Algorithm

QGA, runtime 10.97 s.

7. Optimization Results

Vessel	Class	Route	Cargo t	kn	Fuel	t	Cost \$	CO2e t	ETA h	Util %
V003	Panamax	Port Hedland - Qin	90,899	10.8	LNG	268	405,928	953	364	100
V004	Supramax	Port Hedland - Qin	42,750	10.8	AMMONIA	418	598,962	92	363	100
V008	Panamax	Port Hedland - Qin	85,649	10.9	AMMONIA	653	895,817	144	362	100
V009	Panamax	Paradip - Singapor	90,000	9.8	MGO	162	274,438	623	237	91
V010	Supramax	Port Hedland - Qin	20,702	10.8	LNG	163	276,092	578	364	43
V013	Capesize	Tubarao - Rotterda	150,000	11.3	MGO	670	1,016,985	2,569	493	98
V015	Panamax	Richards Bay - Par	71,300	11.1	LNG	322	492,153	1,142	450	100
V016	Supramax	Richards Bay - Par	48,700	11.1	LNG	257	395,721	912	451	91

8 vessels deployed, 8 held idle.

8-10. Fuel, Cost and Emission Analysis

Metric	Baseline	Optimized	Change %	Direction
Fuel consumption (t)	3,501.1	2,913.3	-16.79%	improved
Operating cost (USD)	4,293,326.5	4,356,096.0	+1.46%	worse
Lifecycle CO2e (t)	12,923.2	7,012.4	-45.74%	improved
Cargo fulfilment (%)	100.0	100.0	+0.00%	worse
Schedule reliability (%)	100.0	100.0	+0.00%	worse

Percentages are computed from the two evaluations above; none are assumed.

11. Pareto Analysis

ID	Cost weight	Cost \$	Lifecycle CO2e t	Cargo %	Pareto-optimal
S00	0.00	4,284,352	7,625	96	yes
S01	0.17	5,572,079	7,393	95	yes
S02	0.33	5,718,582	7,712	97	no

ID	Cost weight	Cost \$	Lifecycle CO2e t	Cargo %	Pareto-optimal
S03	0.50	4,780,231	7,515	92	yes
S04	0.67	4,764,934	8,093	100	no
S05	0.83	6,326,365	7,257	99	yes
S06	1.00	4,681,239	8,303	100	no

4 of 7 sampled configurations are non-dominated. Method: Weighted-sum scalarisation sweep with non-dominated filtering.

12. Benchmarking Results

Algorithm	Runs	Mean fitness	Std dev	Best	Worst	Iter. to converge	Runtime s
QGA	3	2.8801	1.3142	1.0475	4.0643	29.7	2.11
GA	3	1.2331	0.3193	1.0032	1.6846	35.3	0.27
QPSO	3	4.7911	0.0651	4.7136	4.8729	36.3	0.40
PSO	3	4.4467	0.0130	4.4292	4.4602	35.0	0.33
GREEDY	1	16.6450	0.0000	16.6450	16.6450	-	0.00

QGA vs GA: better mean fitness achieved by **GA** (gap +133.57%).

QPSO vs PSO: better mean fitness achieved by **PSO** (gap +7.74%).

Independent runs with distinct random seeds. Mean and standard deviation are reported because metaheuristic outcomes vary by run. No result is asserted in advance of the experiment.

13. Alternative Fuel Comparison

Fuel	\$/t	Annual t	Annual cost \$	Lifecycle CO ₂ e t	vs base %	Payback y
HFO	520	7,478	6,683,063	27,729	+0.0	n/a
MGO	720	7,037	7,773,835	26,994	-2.6	n/a
LNG	640	6,267	6,078,680	22,247	-19.8	19.9
METHANOL	980	15,106	17,296,186	23,036	-16.9	n/a
AMMONIA	1,150	16,160	18,584,296	3,555	-87.2	n/a
HYDROGEN	4,200	2,505	10,521,741	1,127	-95.9	n/a

Transport work is held constant across fuels. Capex and prices are configurable demo parameters, not quotations.

14. Compliance Analysis

Target	Required %	Achieved %	Current g/t-nm	Target g/t-nm	Status
IMO_2030	40	70.5	2.297	4.680	Compliant
IMO_2040	70	70.5	2.297	2.340	Compliant
IMO_2050	100	70.5	2.297	0.000	Above Target

EU ETS estimate for 2026: 2,722 t of allowances at a 100% phase-in and 50% voyage coverage, costing approximately **\$326,677**.

Model estimate from configurable parameters. Not a regulatory calculation, certification or legal advice. Actual compliance depends on audited operational data and the rules in force for the relevant compliance year.

15. Recommendations

1. Adopt the optimised deployment plan: it reduces fuel burn by 16.8% (3,501 t to 2,913 t) against the greedy baseline.
2. Emission reduction of 45.7% is achievable without reducing cargo fulfilment below 100%.
3. The selected plan satisfies every declared constraint and is operationally feasible.
4. On this scenario over 3 independent runs, GA achieved the better mean fitness in QGA vs GA (gap 133.57%).

- 5. On this scenario over 3 independent runs, PSO achieved the better mean fitness in QPSO vs PSO (gap 7.74%).
- 6. No alternative fuel pays back within the modelled horizon at current prices and carbon cost. Revisit if the carbon price rises materially.
- 7. IMO 2050 - net-zero ambition: currently 70.5% reduced against a required 100%. The remaining gap is 2.297 g CO2e per tonne-nm.

16. Methodology Notes

Fuel consumption follows a cubic speed-power relationship with a load-dependent specific fuel oil consumption curve minimised near 75% MCR, corrected for cargo displacement, hull fouling with age, and added resistance from wind, waves and current. Emissions are split well-to-tank and tank-to-wake; lifecycle CO2e is their sum. The optimiser fitness is a weighted sum of normalised fuel, cost, emission and reliability terms plus a static penalty on normalised constraint violation.

17. Dataset Information

Demo dataset generated for simulation and algorithm validation. 6,000 synthetic voyages produced from the documented physical model with injected flow-meter, per-hull and weather noise. This is not real shipping-company data and must not be presented as such.

18. Report Metadata

Generated	2026-09-20 23:19:48
Platform	GreenFleet Quantum
Problem statement	PS-138
Scenario	Case Study 2 - Green Fleet
Algorithm	QGA
Compute	Classical hardware; quantum-inspired algorithms only